

Report on Collision Detection, Isolation and Localization with Base F/T Sensor*

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1 Introduction

The following document reports the methodology used and the results obtained from the simulation on the 3R spatial robot (elbow-type) to estimate point-wise (pure forces) collisions using an extended momentum-based disturbance observer and a base 6D force/torque sensor.

2 Preliminaries

2.1 Dynamic Model

The manipulator used is a fully actuated 3R spatial robot¹ (elbow-type) in the vertical plane; it has three rotational joints: the first one rotates around the same axis where the gravity acts, while the last two rotational axes are parallel and form an angle of 90 degrees with the z_0 -axis of the base reference frame around the x_0 -axis, as shown in Figure 1.

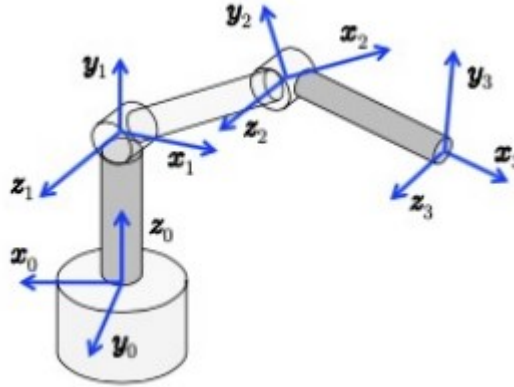


Figure 1: 3R spatial robot (elbow-type) with reference frames

The generalized coordinates vector is $q = (q_1, q_2, q_3)$, respectively the first joint angle w.r.t z_0 -axis, the second joint angle w.r.t z_1 -axis and the third joint angle w.r.t z_2 -axis.

To better understand the kinematics of the robot used, Denavit–Hartenberg parameters are shown in Table 1.

¹Center of masses are on link axes, inertia matrices are diagonal and gravity is present along z_0

Table 1: Denavit–Hartenberg parameters of 3R Spatial Robot

i	α_i	d_i	a_i	θ_i
1	$\pi/2$	L_1	0	q_1
2	0	0	L_2	q_2
3	0	0	L_3	q_3

For the i -th link, denote by m_i , L_i , dc_i and Ii^2 respectively its mass, its length, the distance from the i -th joint axis to its center of mass, and the link inertia moment matrix. Following the Euler-Lagrange approach, the nonlinear dynamic equations of the 3R spatial robot are obtained in the classical form

$$\mathbf{M}(\mathbf{q}) + \mathbf{c}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau} + \boldsymbol{\tau}^{ext} \quad (1)$$

where $\boldsymbol{\tau} = (\tau_1, \tau_2, \tau_3)$ is the torques vector, hereinafter also called covector of joint torques, while the inertia matrix $\mathbf{M}(\mathbf{q})$, the Coriolis and centrifugal forces $\mathbf{c}(\mathbf{q}, \dot{\mathbf{q}})$ and the gravitational forces $\mathbf{g}(\mathbf{q})$ are given as

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix} \quad \mathbf{c}(\mathbf{q}, \dot{\mathbf{q}}) = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \quad \mathbf{g}(\mathbf{q}) = \begin{bmatrix} g_1 \\ g_2 \\ g_3 \end{bmatrix} \quad (2)$$

inertial matrix components as

$$\begin{aligned} m_{11} &= a_1 + a_4 * \cos(q_3) + a_4 * \cos(2 * q_2 + q_3) + a_3 * \cos(q_2 + q_3)^2 + a_2 * \cos(q_2)^2 \\ m_{12} &= m_{13} = m_{21} = m_{31} = 0 \\ m_{22} &= a_5 + 2 * a_4 * \cos(q_3) \\ m_{23} &= a_6 + a_4 * \cos(q_3) \\ m_{32} &= a_6 + a_4 * \cos(q_3) \\ m_{33} &= a_6 \end{aligned} \quad (3)$$

Coriolis and centrifugal vector components as

$$\begin{aligned} c_1 &= -\dot{q}_1 * (\dot{q}_2 * (a_4 * \sin(2 * q_2 + q_3) + a_3 * \cos(q_2 + q_3) * \sin(q_2 + q_3) + \\ &\quad + a_2 * \cos(q_2) * \sin(q_2)) + \dot{q}_3 * ((a_4 * \sin(q_3))/2 + \\ &\quad + (a_4 * \sin(2 * q_2 + q_3))/2 + a_3 * \cos(q_2 + q_3) * \sin(q_2 + q_3))) + \\ &\quad - \dot{q}_1 * \dot{q}_3 * ((a_4 * \sin(q_3))/2 + (a_4 * \sin(2 * q_2 + q_3))/2 + \\ &\quad + a_3 * \cos(q_2 + q_3) * \sin(q_2 + q_3)) - \dot{q}_1 * \dot{q}_2 * (a_4 * \sin(2 * q_2 + q_3) + \\ &\quad + a_3 * \cos(q_2 + q_3) * \sin(q_2 + q_3) + a_2 * \cos(q_2) * \sin(q_2)) \end{aligned}$$

²All these coefficients are positive.

$$\begin{aligned}
c_2 = & (a_4 * \sin(2 * q_2 + q_3) + a_3 * \cos(q_2 + q_3) * \sin(q_2 + q_3) + \\
& + a_2 * \cos(q_2) * \sin(q_2)) * \dot{q}_1^2 - \dot{q}_3 * (a_4 * \dot{q}_2 * \sin(q_3) + \\
& + a_4 * \dot{q}_3 * \sin(q_3)) - a_4 * \dot{q}_2 * \dot{q}_3 * \sin(q_3) \\
c_3 = & ((a_4 * \sin(q_3))/2 + (a_4 * \sin(2 * q_2 + q_3))/2 + \\
& + a_3 * \cos(q_2 + q_3) * \sin(q_2 + q_3)) * \dot{q}_1^2 + a_4 * \sin(q_3) * \dot{q}_2^2
\end{aligned} \tag{4}$$

gravitational vector components as

$$\begin{aligned}
g_1 = & 0 \\
g_2 = & a_8 * \cos(q_2 + q_3) + a_7 * \cos(q_2) \\
g_3 = & a_8 * \cos(q_2 + q_3)
\end{aligned} \tag{5}$$

and setting

$$\begin{aligned}
a_1 = & I_{1zz} + I_{2xx} + I_{3xx} \\
a_2 = & I_{2yy} + m_2 * dc_2^2 + m_3 * L_2^2 - I_{2xx} \\
a_3 = & I_{3yy} + m_3 * dc_3^2 - I_{3xx} \\
a_4 = & m_3 * L_2 * dc_3 \\
a_5 = & I_{2yy} + m_2 * dc_2^2 + m_3 * dc_3^2 + I_{3yy} + m_3 * L_2^2 \\
a_6 = & I_{3yy} + m_3 * dc_3^2 \\
a_7 = & (m_2 * dc_2 + m_3 * L_2) * g_0 \\
a_8 = & m_3 * dc_3 * g_0
\end{aligned} \tag{6}$$

where g_0 is the gravity acceleration. As regards $\boldsymbol{\tau}^{ext} \in R^3$, torque due to the external force $\boldsymbol{F}^{ext} \in R^6$, it is related to the latter by the geometric contact Jacobian $\boldsymbol{J}_c(\boldsymbol{q}) \in R^{6 \times 3}$ according to

$$\boldsymbol{\tau}^{ext} = \boldsymbol{J}_c(\boldsymbol{q})^T \boldsymbol{F}^{ext} \tag{7}$$

The last equation maps the external wrench into external joint torques. In general, even assuming that $\boldsymbol{J}_c(\boldsymbol{q})$ is known, an estimate of $\hat{\boldsymbol{F}}^{ext} \in R^6$ through the pseudo-inverse $\boldsymbol{J}_c(\boldsymbol{q})^\#$ is often not fully reconstructible due to two main reasons (kinematic singularities and collision below the 6th joint) that lead to a reduction in the rank of the contact Jacobian $\text{rank}(\boldsymbol{J}_c(\boldsymbol{q})) < 6$. In our case, the rank of the contact Jacobian is definitely less than 6 because the manipulator consists of only 3 links. For this reason, it is necessary to extend the dynamic model through sensory redundancy, which potentially improves the estimation capability because it allows for more information between the forces/torques exchanged between the robot and the environment.

2.2 Extended Dynamic Model

Additional sensors, which can be mounted at the base, on the end-effector, or along the kinematic structure, are needed to enrich the information for external force estimation. In this work, a 6D F/T sensor is used at the base. The physical presence of the additional sensor can be considered, in terms of the dynamic model, by introducing the concept of virtual joints. The following shows how the dynamic equations of motion, seen in the previous section, are modified to include the virtual joints related to the sensor at the base. The whole structure, the 3R manipulator and the sensor at the base, can be described by the dynamic model through

$$\begin{aligned} \bar{M}(\bar{q}) + \bar{c}(\bar{q}, \dot{\bar{q}}) + \bar{g}(\bar{q}) &= \mathbf{H}\boldsymbol{\tau} + \bar{\boldsymbol{\tau}}^{ext} + \mathbf{A}(\bar{q})^T \boldsymbol{\lambda} \\ \mathbf{A}(\bar{q})\dot{\bar{q}} &= \mathbf{0} \end{aligned} \quad (8)$$

with $\bar{q} = (q_b, q) \in R^{\bar{n}}$ representing the generalized joint positions with $\bar{n} = 6+3$ as the number of extended coordinates to account the physical presence of base sensor. The virtual joints at the base are considered *mechanically locked*, so a velocity constraint $\mathbf{h}(\bar{q}) \in R^6$ and consequently a Jacobian constraint matrix $\mathbf{A}(\bar{q}) \in R^{6 \times 9}$ are expressed in the dynamic model where

$$\mathbf{h}(\bar{q}) = \begin{bmatrix} q_{b1} \\ q_{b2} \\ q_{b3} \\ q_{b4} \\ q_{b5} \\ q_{b6} \end{bmatrix} = \mathbf{0} \quad \mathbf{A}(\bar{q})\dot{\bar{q}} = \frac{\partial \mathbf{h}(\bar{q})}{\partial \bar{q}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \quad (9)$$

and considering $\mathbf{A}(\bar{q})$ is a constant matrix, $\boldsymbol{\lambda} \in R^6$, which is the set of Lagrange multipliers, it represents joint forces and torques acting against the constraints. Hence we can define

$$\bar{\boldsymbol{\tau}}^{ext} = \begin{bmatrix} \boldsymbol{\tau}_b^{ext} \\ \boldsymbol{\tau}^{ext} \end{bmatrix} \quad \boldsymbol{\tau}_b^{ext} = \begin{bmatrix} \mathbf{f}_b^{ext} \\ \mathbf{m}_b^{ext} \end{bmatrix} \quad (10)$$

where $\mathbf{f}_b^{ext}$ and $\mathbf{m}_b^{ext}$ are e force and torque components exerted at the base, respectively. The term $\mathbf{H} \in R^{9 \times 3}$ maps actuation forces to generalized forces acting collocated to $\bar{q}$

$$\mathbf{H} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (11)$$

The overall inertia matrix $\bar{M}(\bar{q}) \in R^{9 \times 9}$, the Coriolis and centrifugal matrix $\bar{c}(\bar{q}, \dot{\bar{q}}) \in R^9$, and the generalized gravity forces $\bar{g}(\bar{q}) \in R^9$ in (8) are extended analogously. In addition, the components of the previous sensor-related inertia matrix may also not be specified because, in the method used in this work for estimating the external forces, momentum is considered, which will remain zero for the virtual joints because, being *mechanically blocked*, the general velocities will remain constantly at zero. For similar reasons, the gravitational terms and the Coriolis and centrifugal forces will be null.

Thus obtaining

$$\bar{M}(\bar{q}) = \begin{bmatrix} *_{6 \times 6} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{6 \times 6} & M(q) \end{bmatrix} \quad \bar{c}(\bar{q}, \dot{\bar{q}}) = \begin{bmatrix} \mathbf{0}_{6 \times 1} \\ c(q, \dot{q}) \end{bmatrix} \quad \bar{g}(\bar{q}) = \begin{bmatrix} \mathbf{0}_{6 \times 1} \\ g(q) \end{bmatrix} \quad (12)$$

2.3 Extended Momentum-Based Disturbance Observer

Utilizing the notion of monitoring system momentum, this approach offers dynamically decoupled residual estimates, leveraging integrated joint torque sensors. It extends the classical momentum-based disturbance observer to accommodate redundancy in force-torque measurements. This work uses a unified framework that integrates all sensor signals to generate an extended residual, eliminating the necessity for acceleration measurements or estimates. The result is a system that provides fully uncoupled dynamics for the external forces and torques obtained. The following are the main steps of the observer used with special reference to the case of the 3R space robot (elbow type). The extended generalized momentum of the robot, encompassing the virtual joints (additional sensors), is formally defined as:

$$\bar{p} = \bar{M}(\bar{q}) \dot{\bar{q}} = \begin{bmatrix} *_{6 \times 6} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{6 \times 6} & M(q) \end{bmatrix} \begin{bmatrix} \mathbf{0}_{6 \times 1} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} \mathbf{0}_{6 \times 1} \\ p \end{bmatrix} \quad (13)$$

as was anticipated in the previous section, the first six components of the vector $\dot{\bar{q}}$ represent the velocities of the virtual joints which will constantly be zero due to mechanical lock. So the first six components of the vector $\bar{p}$ will also be constantly zero, independent on the value of the sensor-related inertia matrix. While the dynamics of the observer will be of the form

$$\dot{\bar{p}} = \bar{\tau} - \bar{n}(\bar{q}, \dot{\bar{q}}) + \bar{r}(t) \quad \dot{\bar{r}} = \bar{K}_0(\dot{\bar{p}} - \dot{\bar{p}}) \quad (14)$$

where

$$\bar{\tau} = \begin{bmatrix} \tau_b \\ \tau \end{bmatrix} \quad \bar{n} = \bar{g} + \bar{c} - \dot{\bar{M}} \dot{\bar{q}} \quad \bar{r}(t) = \bar{K}_0 \left(\bar{p}(t) - \int_0^t \bar{\tau} - \bar{n} + \bar{r} dt - \bar{p}(0) \right) \quad (15)$$

The variables τ_b , represent the base measurements given by the sensor, and τ is the covector of joint torques. Observe that the dependencies on the states have been omitted for the sake of readability. Given a precise model of the dynamics, the residual dynamics can be articulated as follows

$$\dot{\bar{r}} = \bar{K}_0(\bar{\tau}^{ext} - \bar{r}) \quad (16)$$

It can be regarded as a stable, linear, decoupled, first-order filter/estimation of the external generalized forces $\bar{\tau}^{ext}$. In the limit case when $\bar{\mathbf{K}}_0 \rightarrow \infty$ then

$$\bar{\mathbf{r}} \approx \bar{\tau}^{ext} \quad (17)$$

In practical terms, the observer gain involves striking a balance between the convergence rate of the residual and the level of noise present in the signal. In this context, there is no requirement for inverting the inertia matrix to estimate external forces based on additional force/torque signals. This leads to a fully decoupled estimation of external forces and wrenches. Moreover, the necessity for joint acceleration estimates is eliminated. Consequently, the momentum-based external force observer framework fully incorporates sensing redundancy, delivering a pure external wrench estimate associated with the additional sensors.

3 Block Diagram for Simulation

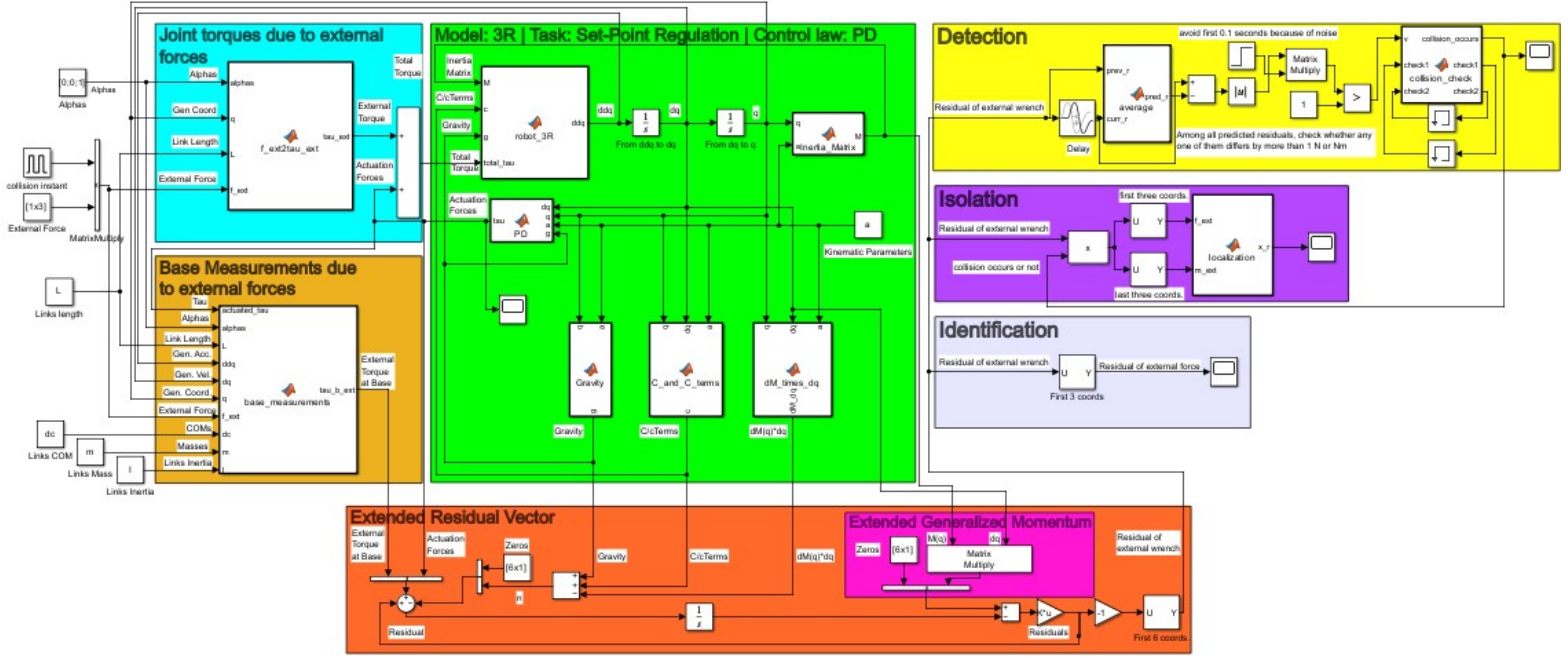


Figure 2: Block diagram to simulate detection, isolation, and localization of external wrench on a 3R spatial robot manipulator (elbow-type)

The following section explains each part that makes up the block diagram by which a collision was simulated with a 3R spatial robot manipulator (elbow-type) equipped with a sensor that detects the forces/torques exerted at the base.

It is specified a priori where the collision occurred and the intensity of the collision along the three axes. The goal is to verify whether or not the last three parts that make up the block diagram (Detection, Isolation and Identification) succeed in reconstructing the point along the robot structure where the collision occurred and the wrench vector using the extended momentum-based observer explained above.

3.1 Joint Torques due to External Forces

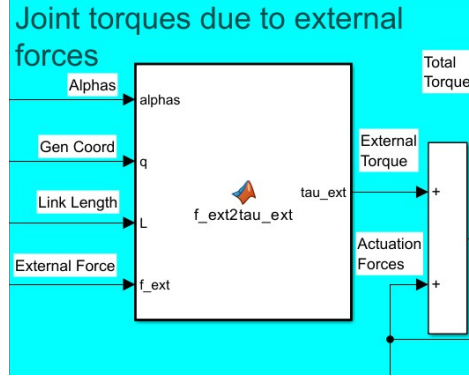


Figure 3: Block diagram part related to joints torques due to external forces

The function $f_ext2tau_ext$ in Figure (3), taking as input the data needed to determine the collision point, constructs the contact Jacobian ³ and through the following formula

$$\tau^{ext} = J_c(q)^T F^{ext}$$

determines τ^{ext} .

The latter vector will then be added to the torques coming out of the control law (PD) to determine the total torques that will regulate the robot joint accelerations under consideration. Since the other input parameters are self-explanatory, it is useful to explain that *Alphas* is a vector of 3 elements and indicates where the collision occurred using the one-hot encoding methodology, with the difference that the value for the link where the collision occurred is a percentage in $(0, 1]$ range. This percentage is applied to the link length to determine the distance between the link joint and the point where the collision occurred.

³Note that the contact Jacobian is determined according to the D-H parameters specified in the Table 1

3.2 Set-Point Regulation with External Force

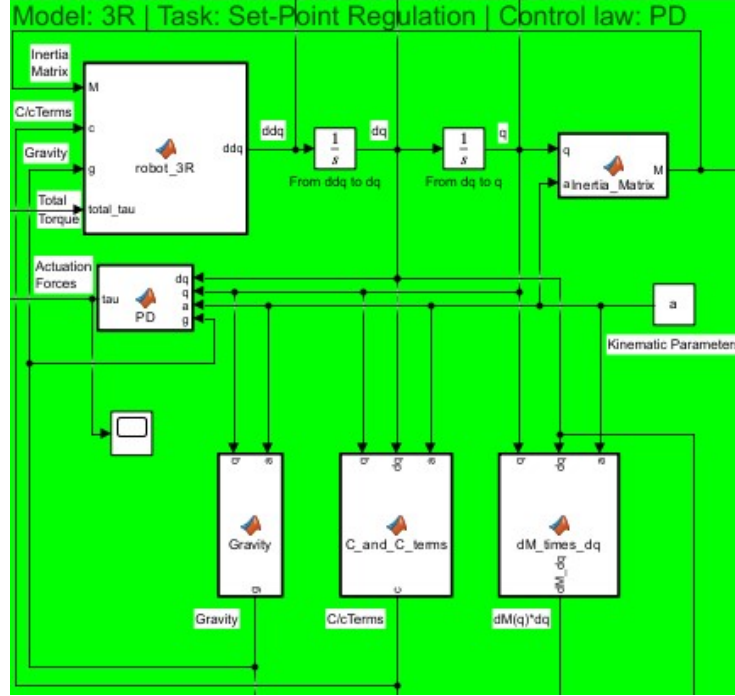


Figure 4: Block diagram part related to the force/torque measurements taken by the sensor at the base

The following part shown in figure (4), simulates joint acceleration⁴ (and consequently velocity and position) using a PD as the control law⁵. Note that the control law used also takes as input the gravitation term to compensate. *Gravity*, *C_and_C_terms* and *dM_times_dq*⁶ are also used to compute $\tilde{n}$ in equation (15)

⁴The Dynamic Model used is the one specified in Section 2

⁵During the completion of this task a collision will occur, in fact, the total torque which determines the movement is given by the output of the previous block diagram part

⁶To be more clear, this function computes the following operation:

$$\dot{M}(\bar{q})\dot{q}$$

3.3 Base Measurements due to External Forces

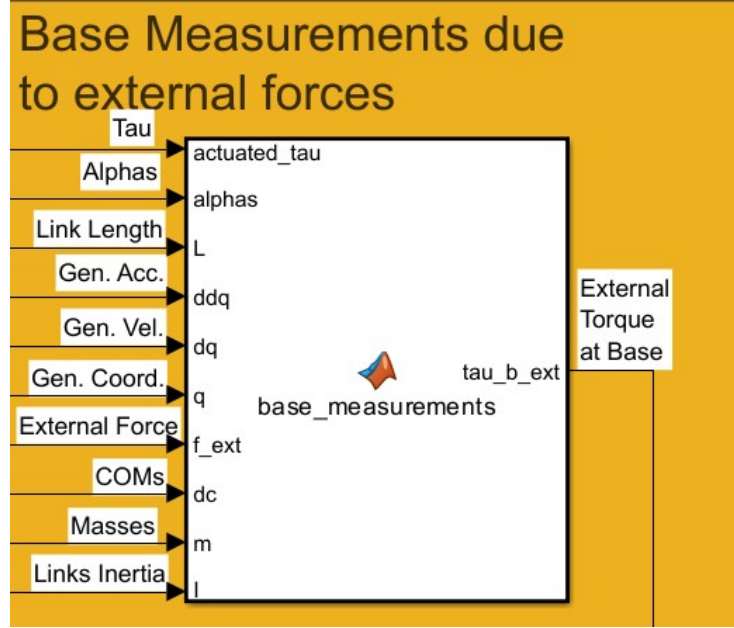


Figure 5: Block diagram part related to the force/torque measurements taken by the sensor at the base

This part simulates the measurements detected by the base F/T sensor. The forces and torques measured by the sensor at the base are those that are calculated from the inverse dynamics using Newton-Euler before projection onto the joint axis (f_0^{ext} and τ_0^{ext} w.r.t RF_0).

3.4 Extended Residual Vector and Extended Generalized Momentum

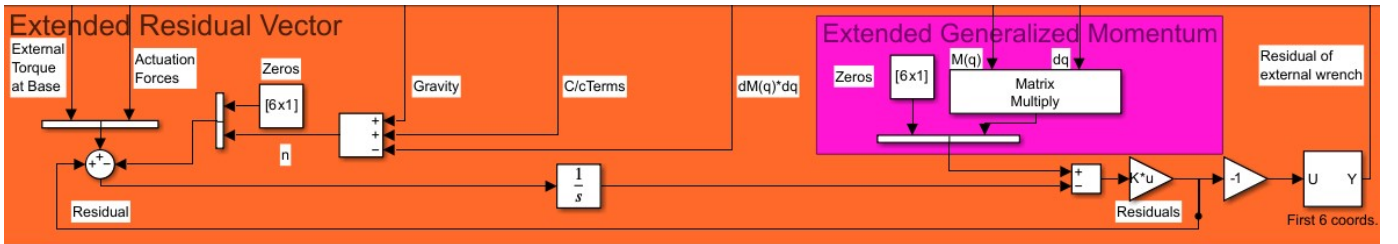


Figure 6: Block diagram part related to Extended Residual Vector and Extended Generalized Momentum

The red part of Figure (6) represents the momentum-based disturbance observer which executes the equation (15) taking into account the equation (13), which is carried out by *Extended Generalized Momentum* part.

3.5 Detection

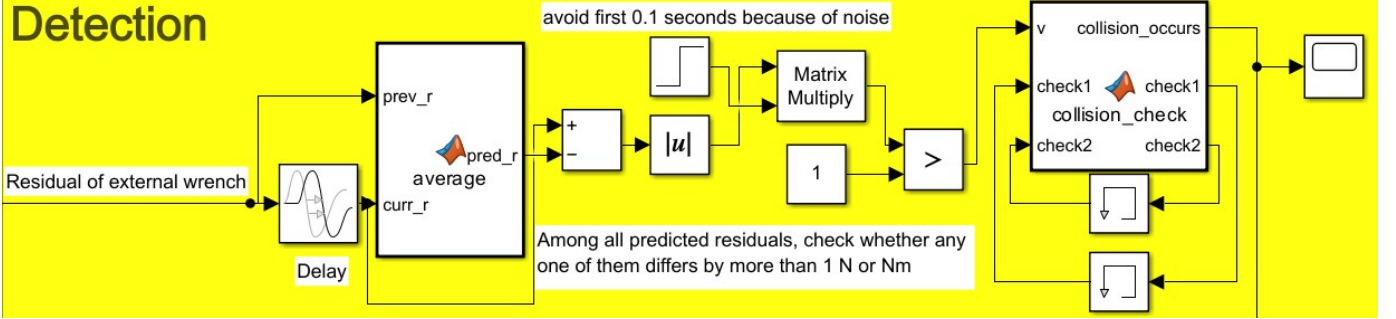


Figure 7: Block diagram part related to the detection phase where it is determined whether or not a collision has occurred

To determine whether or not there has been a collision, the average between the current residual value and the previous one is calculated, then the difference with the current residual value is taken. If this difference exceeds 1N (or 1Nm) then a collision has occurred. This value can be changed based on sensor accuracy and available sampling time. Note that a step signal is also used to avoid considering the data relating to the first moments of the experiment which contain a lot of noise. The averaging function can be seen as a kind of interpolation that tries to predict the value of the current residual. In fact, during normal operation of the manipulator, in the absence of collisions the difference between the current residual and the predicted residual remains below the threshold (also for this reason the absolute function was used).

Finally, the *collision_check* in Figure (7) function raises a flag if, in at least one of the six residuals, a collision has been detected. The two *check* variables respectively have the following tasks: to signal that a collision has already occurred by keeping the flag raised and to avoid the end of the collision as a new collision⁷.

⁷The preceding *collision_check* function will succeed on two occasions: when the collision begins and when it ends

3.6 Isolation

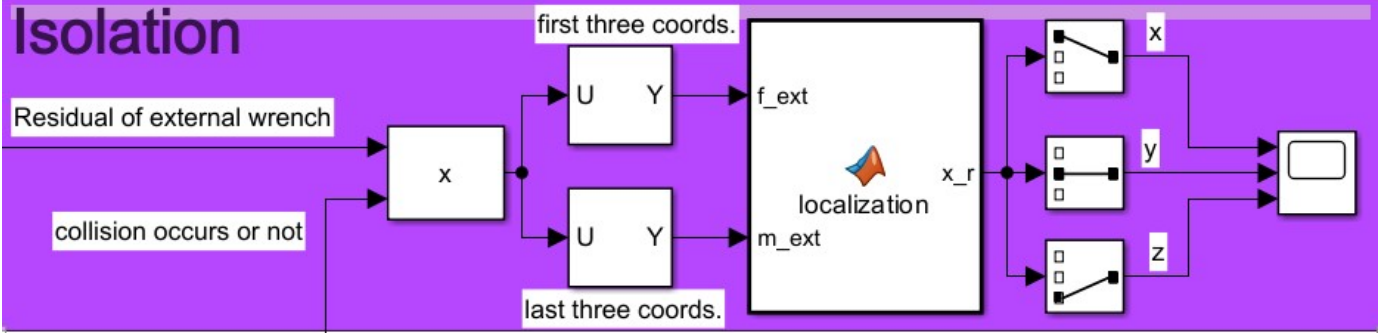


Figure 8: Block diagram part related to the isolation, or localization, phase where it is determined where a collision has occurred

The residual of the external wrench ($\bar{\mathbf{r}}$), or the first six components of the vector $\bar{\mathbf{r}}$, are multiplied with the output of the *collision_check* function (in Figure (7)), which will return 0 in the absence of collision, 1 otherwise. The *isolation* function in Figure (8) returns the point where the collision occurred, via three coordinates, executing the following formula

$$\mathbf{x}_r = -\mathbf{S}(\mathbf{f}_b^{ext})\#\boldsymbol{\tau}_b^{ext} \quad (18)$$

where $\mathbf{S}()$ is the skew-symmetric matrix operator for the vector product, $\#$ denotes a generalized pseudo-inverse and for the equations (16) and (17) $\mathbf{f}_b^{ext}$ is given by the first three coordinates of external wrench residual ($\bar{\mathbf{r}}$), while $\boldsymbol{\tau}_b^{ext}$ is given by the last three coordinates.

3.7 Identification

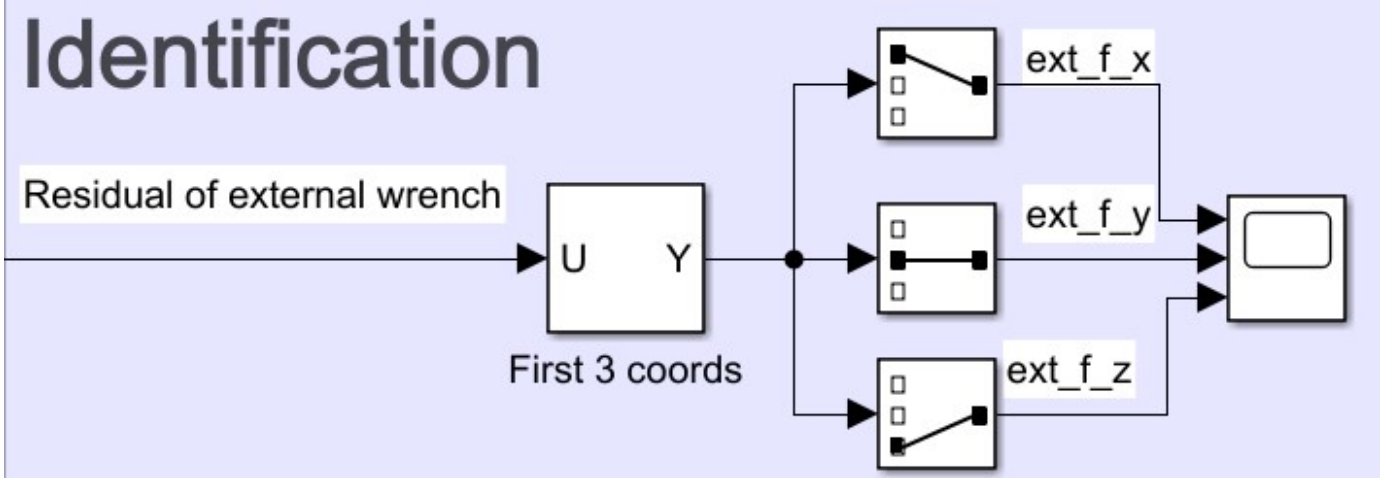


Figure 9: Block diagram part related to the identification phase where it is determined the external force/torque magnitude along the three axes

Finally, regarding the identification phase, it has already been implicitly performed in the isolation phase. Because, it is necessary to regulate the external force/torque vector to determine the collision point and perform the *solution* function, but here it is shown for greater clarity. Since it was assumed that we could only observe point collisions (pure forces), only the first three components of $(\bar{\mathbf{r}})$, i.e. τ_b^{ext} , are taken.

4 Simulation Result

4.1 Detection Phase

Table 2: Simulation Parameters

Duration [s]	Collision [s]	Hit Link	$\mathbf{q}$	Collision Point	External Force [N]
3	1.8	3	$\begin{bmatrix} 0 \\ \pi/2 \\ 0 \end{bmatrix}$	$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1.4 \end{bmatrix}$	$\begin{bmatrix} f_x^{ext} \\ f_y^{ext} \\ f_z^{ext} \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \\ 0 \end{bmatrix}$

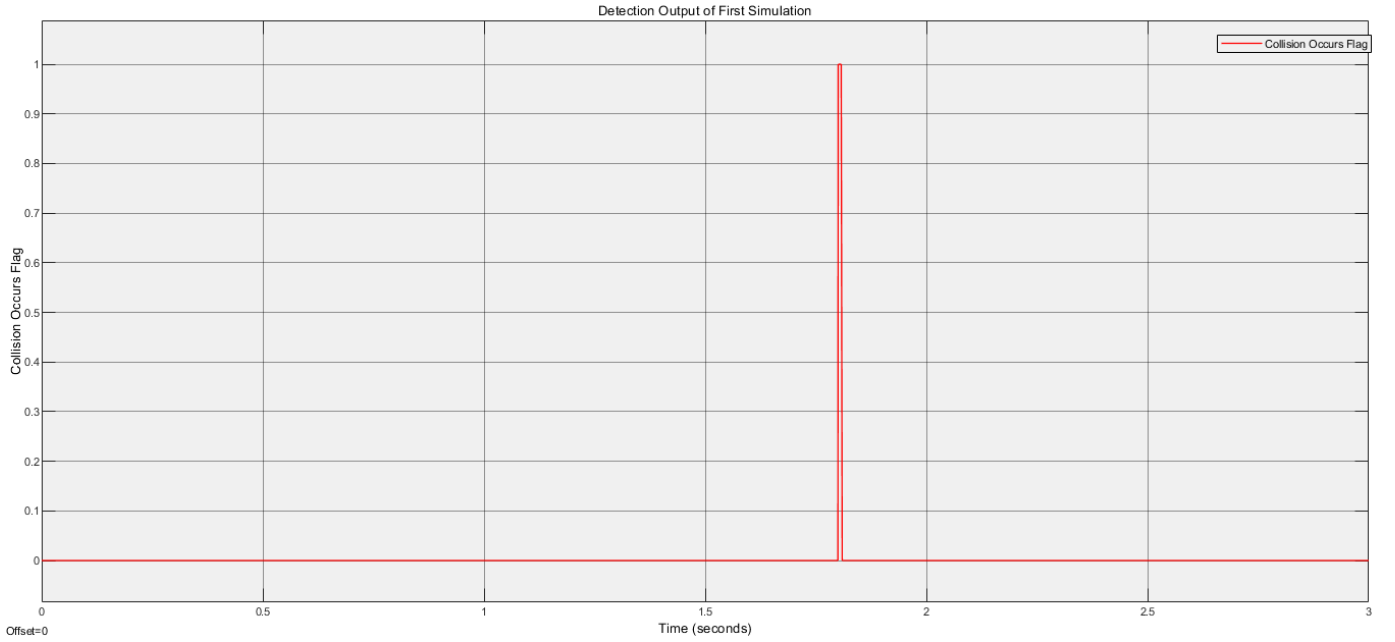


Figure 10: Detection Output

As shown in Figure (10), the part of the block diagram relating to the detection phase detects a collision 1.8 seconds after the start of the simulation, as expected.

4.2 Isolation Phase

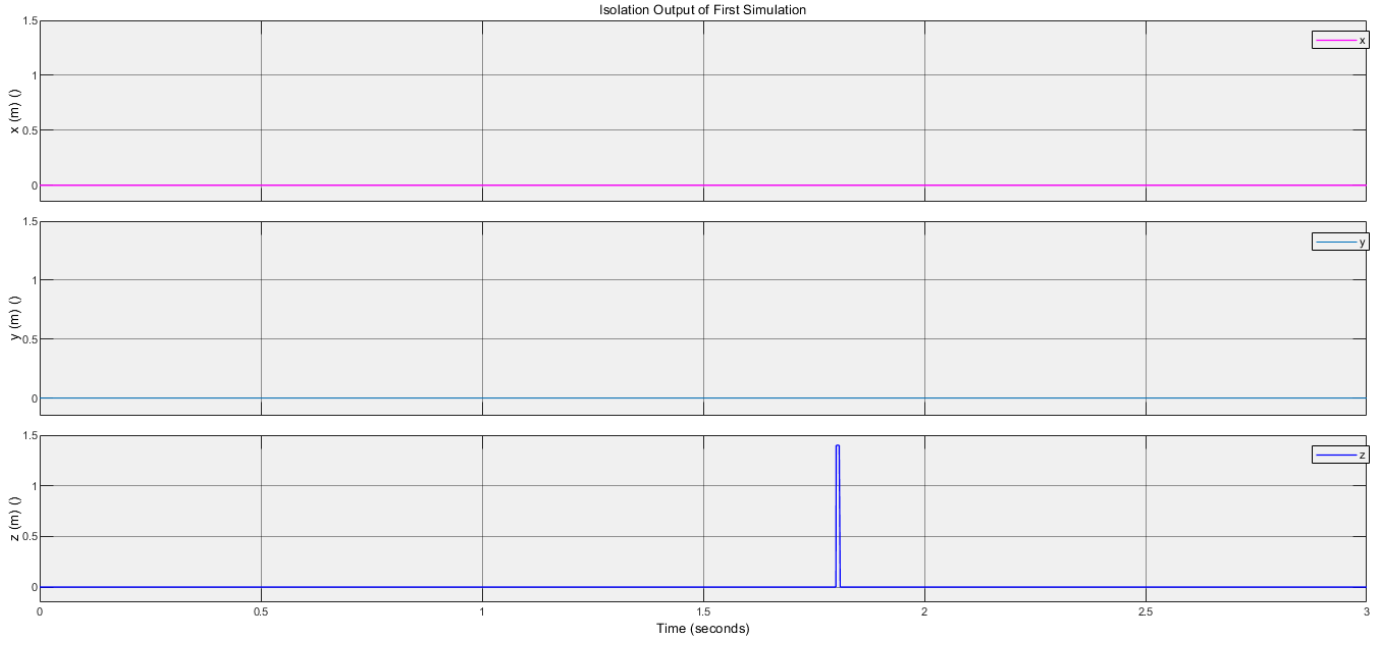


Figure 11: Isolation Output

As shown in Figure (11), the point where a collision is detected is $[x, y, z] = [0, 0, 1.4]$, as expected, and the accurate contact location can be determined when there is an assumption of possessing comprehensive knowledge about the structure's geometry.

4.3 Identification Phase

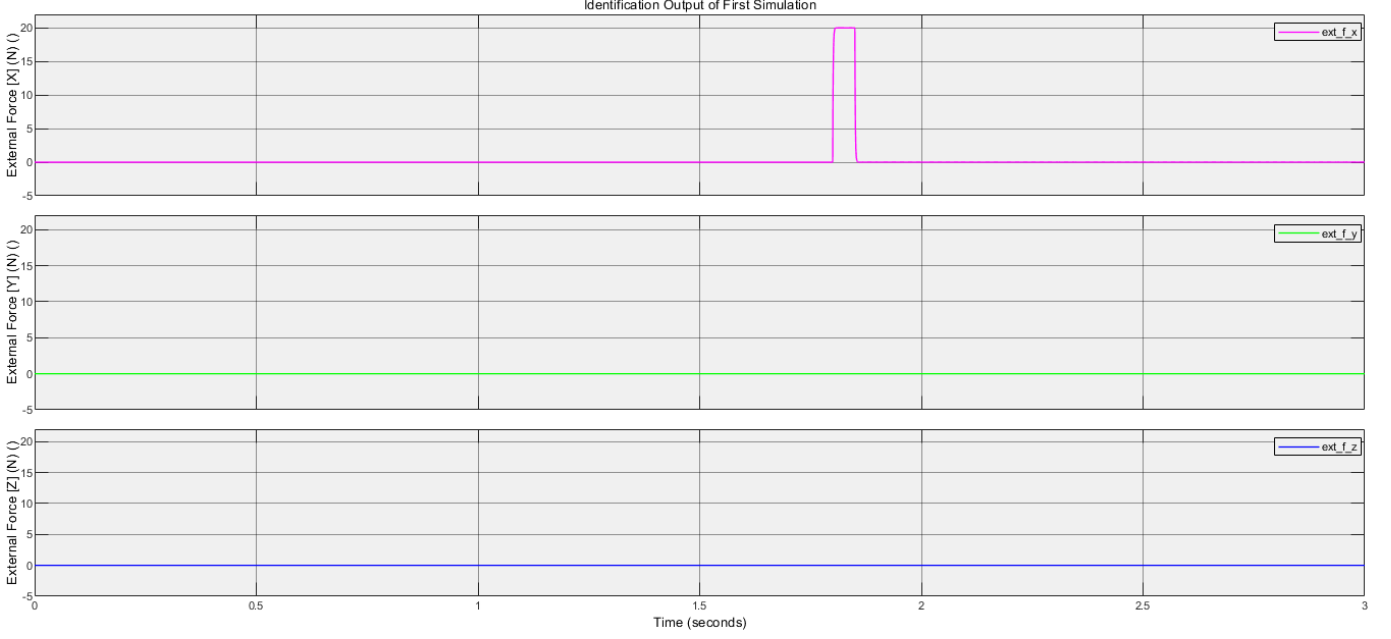


Figure 12: Identification Output

As shown in Figure (12) the estimated/detected external force during the collision is $[f_x^{ext}, f_y^{ext}, f_z^{ext}] = [20, 0, 0]$, as expected.

5 Conclusions

A cohesive framework based on momentum has been used to estimate point-wise (pure forces) collisions on 3R spatial robot (elbow-type). As was shown, decoupled force-torque estimations in Cartesian space were achieved through the use of an additional base 6D F/T sensor. This independence from joint configuration and immunity to singularities enhance the reliability of the estimates. Furthermore, an experiment was conducted in which the external force was estimated with high accuracy and in which the collision point was found despite the robot being in a singularity and the external force causing additional external torques for some joints (i.e. third joint).